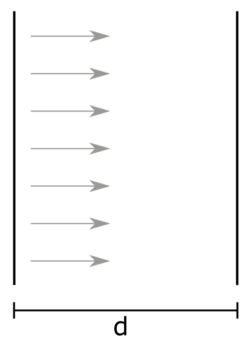
Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Relativistic Diode 20 points

Calculate the current density J for an extreme relativistic 1D diode. (i.e. ignore the small region of the diode for which $\beta \simeq 1$ is not a good approximation.) Assume space-charge limited emission. Let the diode have length d, and voltage V, and the ion species have mass m and charge q. Sketch $\log J$ vs. $\log V$ for both a non-relativistic diode and an extreme relativistic diode and comment on how the two cases scale.



Hint:

$$\nabla \cdot E = \frac{\rho}{\epsilon} \quad \text{and} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$$

are both relativistically correct equations.

Problem 2. FRIB Solenoids 15 pts.

FRIB will use superconducting solenoids for beam focusing in the three linac segments. Consider U_{238}^{+78} (post stripper) accelerated to a kinetic energy of 128 MeV/u. The solenoids have an axial length of approximately 0.5 m and a field strength of 7 T.

a) 5 pts: Estimate $[B\rho]$ for the ion in T-m.

b) 5 pts: Estimate the focal length, f, of the solenoid assuming $B_z \approx 7$ T over the axial length and is zero outside (hard edge solenoid). Is the thin lens approximation reasonable? Why?

c) 5 pts: Suppose superconducting cable with an average current density of J is wound to a thickness T around the beam pipe. Derive a formula to roughly estimate the central field, B_z in terms of J and T. If the coil thickness is $T = 2$ cm what is the required current density in A/mm^2 to produce B_z=7 T?

Problem 3. Magnetic chicane

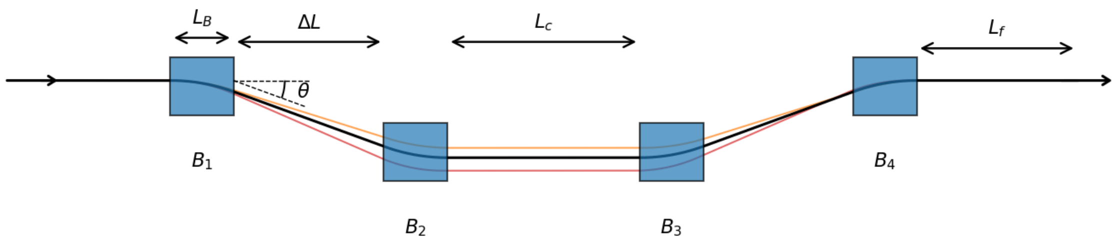


Figure 1: Magnetic chicane

Bunch compression of an electron beam is achieved by making leading electrons propagate slower and trailing electrons propagate faster, hence shortening the bunch length. In the relativistic regime where electrons travel in the speed of approaching the speed of light, energy-dependent path-length-difference is induced to make leading (trailing) electrons travel a longer (shorter) path to achieve bunch compression. A magnetic chicane is a type of bunch compressor that employs four dipole magnets arranged in the form shown in Fig. 1.

a) 1 pts: The straight-through path (if no dipoles were present) has length:

$$s_{\text{straight}} = 4L_B + 2\Delta L + L_c + L_f \ .$$

The path travelled by an electron beam passing through a magnetic chicane is shown as the black trajectory line shown in Fig. 1. Find the expression of path length $s(\theta)$.

b) 5 pts: Find the path-length difference, $\Delta s = s(\theta) - s_{\text{straight}}$, do Taylor expansion up to the second order and show that

$$\Delta s = s(\theta) - s_{\text{straight}} \approx \theta^2(\Delta L + \frac{2}{3}L_B) \ . \tag{1}$$

c) 6 pts: The path length difference can be expressed as

$$\Delta s = \Delta s_0 + \mathrm{R}_{56}\delta + \mathrm{T}_{566}\delta^2 + \mathrm{U}_{5666}\delta^3 + \mathcal{O}(\delta^4) \ , \tag{2}$$

where Δs_0 is the path-length difference of the designed energy E_0. Plug in $\theta = \theta_0/(1+\delta)$ into eq. 1, then expand it up to the third order, then compare it with eq. 2 to show that

$$\mathrm{R}_{56} \approx -2\theta_0^2(\Delta L + \frac{2}{3}L_B) \ ,$$
$$\mathrm{T}_{566} \approx -\frac{3}{2}\mathrm{R}_{56},$$
$$\mathrm{U}_{5666} \approx 2\mathrm{R}_{56}.$$

d) 3 pts: Check Fig. 2. This FEL has chicane bunch compressors that compress beams from (a) to (b). Beams get worse from (b) to (c) due to wakefields. After that, a special bunch compressor compress beams from (c) to (d). Does this special bunch compressor need $\mathrm{R}_{56} > 0$ or $\mathrm{R}_{56} < 0$? Why?

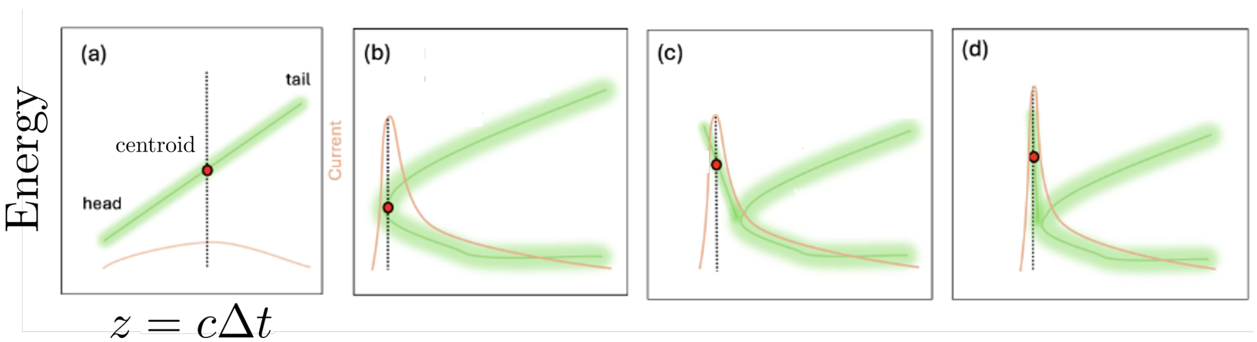


Figure 2: Longitudinal phase space at (a) before chicane, (b) after chicane, (c) before a special bunch compressor, (d) final outcome.